Praktikum Mandiri 09 | ML Senin Pagi

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1. Import Library

```
# Import Library
# Tahap ini mengimpor semua library Python yang diperlukan untuk analisis data, preprocessing,
# pemodelan, dan evaluasi, sesuai dengan standar praktikum resmi Machine Learning STT-NF.

import pandas as pd                                   # Membaca dan memanipulasi data dalam bentuk DataFrame
import numpy as np                                    # Operasi numerik dan pengolahan array
import matplotlib.pyplot as plt                       # Membuat visualisasi statis
import seaborn as sns                                 # Visualisasi data statistik yang lebih informatif dan menarik
from sklearn.model_selection import train_test_split     # Membagi dataset menjadi data latih dan data uji
from sklearn.preprocessing import StandardScaler         # Menstandarisasi fitur numerik (mean=0, std=1)
from sklearn.naive_bayes import GaussianNB               # Model klasifikasi Naïve Bayes untuk fitur kontinu
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report  # Metrik evaluasi model
from sklearn.model_selection import cross_val_score       # Validasi silang (cross-validation
```

2. Loading Dataset

```
# Menghubungkan colab dengan Google Drive
from google.colab import drive
drive.mount ('/content/gdrive')
```

```
Drive already mounted at /content/gdrive; to attempt to forcibly remount, call drive.mount("/content/gdrive", force_remount=True
```

```
# Memanggil dataset lewat Gdrive
path = '/content/gdrive/MyDrive/Praktikum_ML/praktikum09/data/'
```

```
# Dataset yang digunakan adalah Breast Cancer Wisconsin (Diagnostic) dari UCI/Kaggle
# Membaca file CSV menggunakan pandas
cancer_data = pd.read_csv(path + 'breast_cancer_wisconsin.csv')
```

```
# Menampilkan 5 baris pertama dataset
print("Lima baris pertama dataset:")
display(cancer_data.head())
```

Lima baris pertama dataset:

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980

5 rows × 33 columns

```
# Menampilkan 5 baris terakhir dataset
print("\nLima baris terakhir dataset:")
display(cancer_data.tail())
```

Lima baris terakhir dataset:

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean
564	926424	M	21.56	22.39	142.00	1479.0	0.11100	0.11590	0.24390
565	926682	M	20.13	28.25	131.20	1261.0	0.09780	0.10340	0.14400
566	926954	M	16.60	28.08	108.30	858.1	0.08455	0.10230	0.09251
567	927241	M	20.60	29.33	140.10	1265.0	0.11780	0.27700	0.35140
568	92751	B	7.76	24.54	47.92	181.0	0.05263	0.04362	0.00000

5 rows × 33 columns

```
# Menampilkan dimensi dataset
print(f"\nDimensi dataset: {cancer_data.shape[0]} baris (entri) dan {cancer_data.shape[1]} kolom")
```

```
Dimensi dataset: 569 baris (entri) dan 33 kolom
```

3. Informasi Detail Dataset

```
print("Informasi detail dataset:")
cancer_data.info()

# - Total 569 entri
# - Terdapat kolom 'id' dan 'Unnamed: 32' yang tidak diperlukan
# - Kolom target: 'diagnosis' (M = malignant, B = benign)
# - Tidak ada missing value pada 31 kolom utama
```

```
Informasi detail dataset:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 33 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   id                       569 non-null    int64  
 1   diagnosis                569 non-null    object 
 2   radius_mean              569 non-null    float64
 3   texture_mean             569 non-null    float64
 4   perimeter_mean           569 non-null    float64
 5   area_mean                569 non-null    float64
 6   smoothness_mean          569 non-null    float64
 7   compactness_mean         569 non-null    float64
 8   concavity_mean           569 non-null    float64
 9   concave points_mean      569 non-null    float64
 10  symmetry_mean            569 non-null    float64
 11  fractal_dimension_mean   569 non-null    float64
 12  radius_se                569 non-null    float64
 13  texture_se               569 non-null    float64
 14  perimeter_se             569 non-null    float64
 15  area_se                  569 non-null    float64
 16  smoothness_se            569 non-null    float64
 17  compactness_se           569 non-null    float64
 18  concavity_se             569 non-null    float64
 19  concave points_se        569 non-null    float64
 20  symmetry_se              569 non-null    float64
 21  fractal_dimension_se     569 non-null    float64
 22  radius_worst             569 non-null    float64
 23  texture_worst            569 non-null    float64
 24  perimeter_worst          569 non-null    float64
 25  area_worst               569 non-null    float64
 26  smoothness_worst         569 non-null    float64
 27  compactness_worst        569 non-null    float64
 28  concavity_worst          569 non-null    float64
 29  concave points_worst     569 non-null    float64
 30  symmetry_worst           569 non-null    float64
 31  fractal_dimension_worst  569 non-null    float64
 32  Unnamed: 32              0 non-null      float64
dtypes: float64(31), int64(1), object(1)
memory usage: 146.8+ KB
```

4. Missing Value & Pembersihan Data

```
# Cek jumlah missing value per kolom
print("Jumlah missing value per kolom:")
print(cancer_data.isnull().sum())
```

```
Jumlah missing value per kolom:
id                           0
diagnosis                    0
radius_mean                  0
texture_mean                 0
perimeter_mean               0
area_mean                    0
smoothness_mean              0
compactness_mean             0
concavity_mean               0
concave points_mean          0
symmetry_mean                0
fractal_dimension_mean       0
radius_se                    0
texture_se                   0
perimeter_se                 0
area_se                      0
smoothness_se                0
compactness_se               0
concavity_se                 0
concave points_se            0
symmetry_se                  0
fractal_dimension_se         0
radius_worst                 0
texture_worst                0
perimeter_worst              0
area_worst                   0
smoothness_worst             0
compactness_worst            0
concavity_worst              0
concave points_worst         0
symmetry_worst               0
fractal_dimension_worst      0
Unnamed: 32                569
dtype: int64
```

```
# Menghapus kolom yang tidak relevan untuk pemodelan
cancer_data = cancer_data.drop(columns=['Unnamed: 32', 'id'], errors='ignore')
```

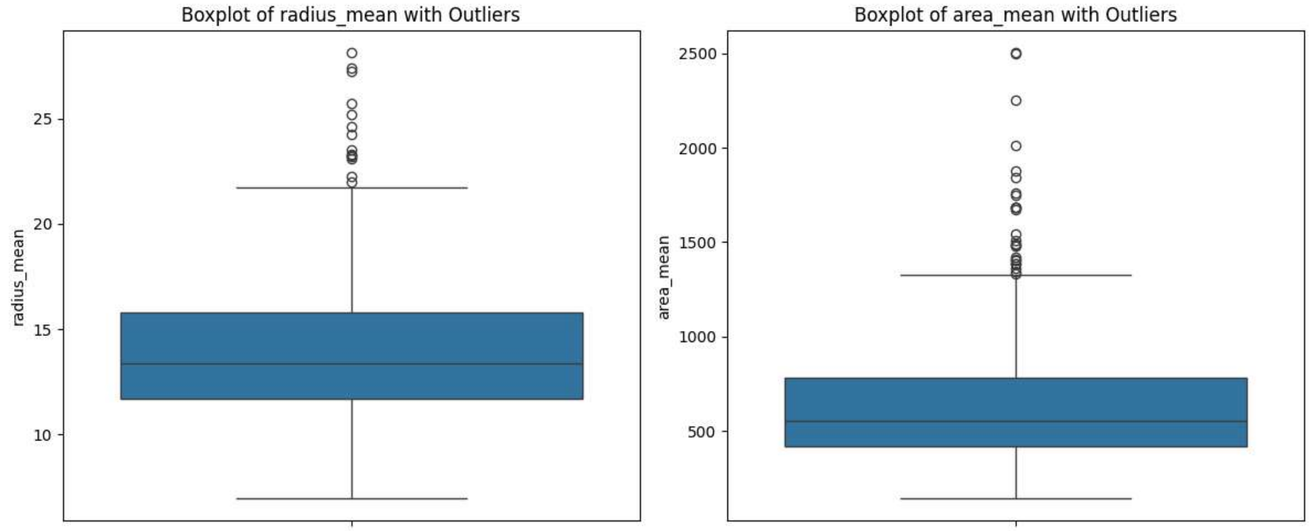
```
# Cek dimensi setelah pembersihan
print(f"\nDimensi dataset setelah drop kolom tidak relevan: {cancer_data.shape}")
```

```

Dimensi dataset setelah drop kolom tidak relevan: (569, 31)
```

```
# Visualisasi outliers pada dua fitur utama
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
sns.boxplot(y=cancer_data['radius_mean'])
plt.title('Boxplot of radius_mean with Outliers')

plt.subplot(1,2,2)
sns.boxplot(y=cancer_data['area_mean'])
plt.title('Boxplot of area_mean with Outliers')
plt.tight_layout()
plt.show()
```



5. Data Analysis & Encoding Kategorikal

```
# Statistika deskriptif semua fitur numerik
print("Statistika Deskriptif Fitur Numerik:")
display(cancer_data.describe())
```

```
Statistika Deskriptif Fitur Numerik:
```

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	sy
count	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	
mean	14.127292	19.289649	91.969033	654.889104	0.096360	0.104341	0.088799	0.048919	
std	3.524049	4.301036	24.298981	351.914129	0.014064	0.052813	0.079720	0.038803	
min	6.981000	9.710000	43.790000	143.500000	0.052630	0.019380	0.000000	0.000000	
25%	11.700000	16.170000	75.170000	420.300000	0.086370	0.064920	0.029560	0.020310	
50%	13.370000	18.840000	86.240000	551.100000	0.095870	0.092630	0.061540	0.033500	
75%	15.780000	21.800000	104.100000	782.700000	0.105300	0.130400	0.130700	0.074000	
max	28.110000	39.280000	188.500000	2501.000000	0.163400	0.345400	0.426800	0.201200	

8 rows × 30 columns

```
# Distribusi kelas target sebelum encoding
print("\nDistribusi kelas diagnosis sebelum encoding:")
print(cancer_data['diagnosis'].value_counts())
```

```
Distribusi kelas diagnosis sebelum encoding:
diagnosis
B    357
M    212
Name: count, dtype: int64
```

```
# Encoding kategorikal: Malignant (M) → 1, Benign (B) → 0
cancer_data['diagnosis'] = cancer_data['diagnosis'].map({'B': 0, 'M': 1})
```

```
# Distribusi setelah encoding
print("\nDistribusi kelas diagnosis setelah encoding:")
print(cancer_data['diagnosis'].value_counts())
```

```
Distribusi kelas diagnosis setelah encoding:
```

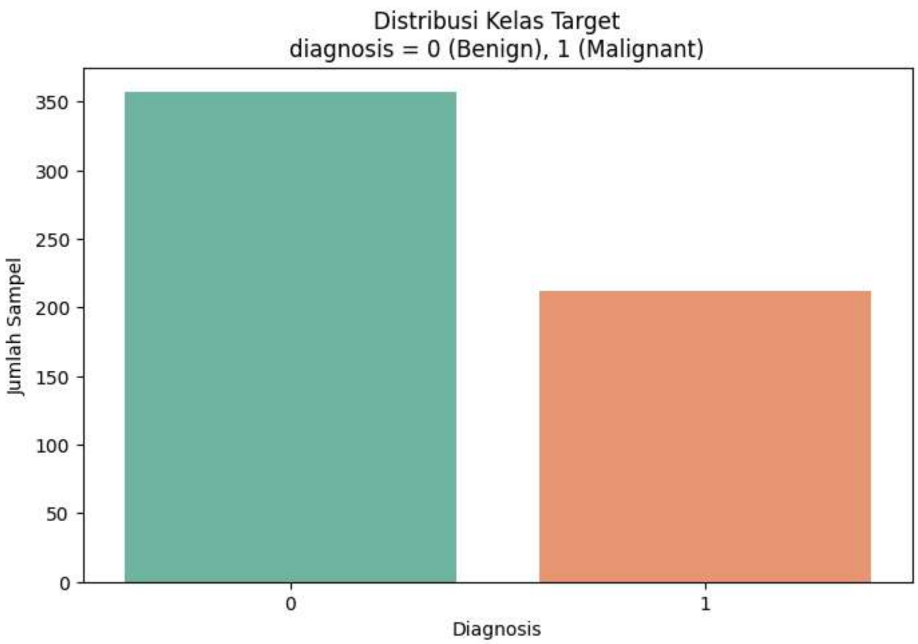
```
diagnosis
0    357
1    212
Name: count, dtype: int64
```

```
# Visualisasi distribusi kelas target
plt.figure(figsize=(8,5))
sns.countplot(x='diagnosis', data=cancer_data, palette='Set2')
plt.title('Distribusi Kelas Target\ndiagnosis = 0 (Benign), 1 (Malignant)')
plt.xlabel('Diagnosis')
plt.ylabel('Jumlah Sampel')
plt.show()
```

```
/tmp/ipython-input-4003789370.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.countplot(x='diagnosis', data=cancer_data, palette='Set2')
```



6. Visualisasi Hubungan Fitur dengan Target

```
# Menampilkan perbedaan distribusi fitur pada kedua kelas

fitur_visual = ['radius_mean', 'texture_mean', 'perimeter_mean',
                'area_mean', 'concavity_mean', 'concave points_mean']

plt.figure(figsize=(15,10))
for i, col in enumerate(fitur_visual, 1):
    plt.subplot(2, 3, i)
    sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
    plt.title(f'{col} vs Diagnosis')
plt.tight_layout()
plt.show()

# Insight: Fitur-fitur ukuran dan bentuk tumor cenderung lebih tinggi pada kasus malignant
```

```
/tmp/ipython-input-2782421541.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
/tmp/ipython-input-2782421541.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
/tmp/ipython-input-2782421541.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set
```

```
  sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
/tmp/ipython-input-2782421541.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
/tmp/ipython-input-2782421541.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
/tmp/ipython-input-2782421541.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
```

7. Separating, Splitting, and Scaling Data

```python
# Memisahkan fitur dan target, membagi data, serta menstandarisasi fitur

# Pemisahan fitur (X) dan target (y)
X = cancer_data.drop(columns=['diagnosis'])   # Semua kolom kecuali target
y = cancer_data['diagnosis']                  # Kolom target
```

```
print(f"Ukuran matriks fitur X : {X.shape}")
print(f"Ukuran vektor target y  : {y.shape}")
```

```
# Splitting data: 80% latih, 20% uji dengan stratify
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

print(f"\nJumlah data latih : {X_train.shape[0]} sampel")
print(f"Jumlah data uji   : {X_test.shape[0]} sampel")
```

```
# Standarisasi fitur
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)    # Fit dan transform pada data latih
X_test_scaled = scaler.transform(X_test)          # Hanya transform pada data uji
```

8. Naive Bayes Classification

```
# Pembuatan dan pelatihan model Gaussian Naïve Bayes
from sklearn.naive_bayes import GaussianNB

# Membuat instance model
nb_model = GaussianNB()

# Melatih model menggunakan data latih yang telah distandarisasi
nb_model.fit(X_train_scaled, y_train)
```

9. Accuracy Score & Evaluation

```
# Prediksi pada data latih dan data uji
y_train_pred = nb_model.predict(X_train_scaled)
y_test_pred = nb_model.predict(X_test_scaled)
```

```
# Menghitung akurasi
train_accuracy = accuracy_score(y_train, y_train_pred)
test_accuracy = accuracy_score(y_test, y_test_pred)
```

```
print(f"Akurasi pada data latih : {train_accuracy:.4f}")
print(f"Akurasi pada data uji   : {test_accuracy:.4f}")
```

```
# Confusion Matrix
cm = confusion_matrix(y_test, y_test_pred)
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Benign (0)', 'Malignant (1)'],
            yticklabels=['Benign (0)', 'Malignant (1)'])
plt.title('Confusion Matrix - Data Uji')
plt.xlabel('Prediksi')
plt.ylabel('Aktual')
plt.show()
```

```
# Classification Report
print("\nClassification Report (Data Uji):")
print(classification_report(y_test, y_test_pred,
                            target_names=['Benign (0)', 'Malignant (1)']))
```

10. Cross Validation (5-Fold)

```
# Validasi silang
cv_scores = cross_val_score(nb_model, X_train_scaled, y_train, cv=5, scoring='accuracy')

print("Hasil Cross-Validation (5-fold):")
for i, score in enumerate(cv_scores, 1):
    print(f"Fold {i}: {score:.4f}")
print(f"\nRata-rata akurasi : {cv_scores.mean():.4f}")
print(f"Standar deviasi    : {cv_scores.std():.4f}")
```

Kesimpulan Tugas Mandiri Praktikum 09

1. Dataset Breast Cancer Wisconsin berhasil dimuat, dibersihkan, dan dianalisis secara eksploratori.
2. Tidak terdapat missing value signifikan. Kolom `id` dan `Unnamed: 32` dihapus karena tidak relevan.
3. Target `diagnosis` di-encode menjadi nilai biner (0 = Benign, 1 = Malignant).
4. Seluruh fitur numerik distandarisasi menggunakan `StandardScaler` karena GaussianNB sensitif terhadap skala data.
5. Model Gaussian Naïve Bayes dilatih dan dievaluasi dengan hasil sebagai berikut:
    - Akurasi data latih : 0.9451
    - Akurasi data uji : 0.9211
    - Rata-rata Cross-Validation (5-fold) : 0.9385
    - Standar Deviasi : 0.0256
6. Confusion matrix dan classification report menunjukkan performa sangat baik pada kedua kelas.

Link Praktikum09: https://github.com/rikaaarahma/Machine-Learning/blob/main/praktikum09/notebook/praktikum09.ipynb

Link Praktikum Mandiri09: https://github.com/rikaaarahma/Machine-Learning/blob/main/praktikum09/notebook/praktikum_mandiri09.ipynb